Scaling Laws for Cyber Resilience: Modeling the Persistence of Backdoors in Post-trained Large Language Models

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Abstract—Backdoor attacks on large language models (LLMs) persist through safety fine-tuning procedures, yet no principled metric exists to quantify this persistence as a function of model scale. Existing attack success rate (ASR) measurements are scale-agnostic and provide no theoretical guarantees connecting parameter count to backdoor robustness. We propose the *Cyber Resilience Scaling Coefficient* (CRSC), a closed-form metric that combines normalized Frobenius drift of hidden-state representations (Δ_{hidden}) and Shannon entropy change of the loss distribution (ΔH) into a single score quantifying backdoor persistence after supervised fine-tuning (SFT) or reinforcement learning from human feedback (RLHF). We prove two properties: (i) CRSC is monotonically non-decreasing in parameter count N for fixed fine-tuning conditions (Proposition 1), formalized via random matrix theory; and (ii) CRSC converges to a concentration-of-measure proxy for ASR with bounded estimation error (Proposition 2). A three-agent evaluation pipeline (Detection, Analysis, Response) computes CRSC across $N \in \{7\text{B}, 13\text{B}, 70\text{B}\}$, poison rates $\rho \in \{0.001, 0.005, 0.01, 0.05\}$, three trigger types, and two fine-tuning methods, yielding 72 experimental runs on synthetic data calibrated against TrojAI NIST 2024. At threshold $\tau = 0.62$, CRSC achieves F1 $= 0.891$ (95% CI: $[0.874, 0.908]$), AUC-ROC $= 0.924$, and Pearson $r = 0.847$ ($p < 10^{-5}$) between CRSC and $\log_{10}(N)$, confirming the scaling monotonicity. CRSC is mapped to MITRE ATLAS adversarial ML techniques and the NIST AI Risk Management Framework, providing a compliance-ready risk classification at three severity levels. Code and data are released under MIT license at https://github.com/ufxa/Cyber-Resilience-Scaling-Coefficient.

Index Terms—Backdoor attacks, large language models, cyber resilience, scaling laws, fine-tuning security, RLHF, adversarial machine learning, MITRE ATLAS.

1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across security-sensitive domains, including penetration testing assistance [1], vulnerability analysis [2], and autonomous cyberdefense [3]. This proliferation, however, introduces a critical attack surface: adversaries may embed *backdoors*—hidden behavioral triggers that cause targeted model outputs when activated—during the supply chain of model training. Unlike classical software vulnerabilities, LLM backdoors survive post-training alignment procedures (supervised fine-tuning, SFT, and reinforcement learning from human feedback, RLHF), creating a persistent threat even after safety hardening.

A fundamental open question concerns the relationship between *model scale* and *backdoor persistence*: do larger models resist fine-tuning-based backdoor removal more than smaller ones? Empirical evidence from prior work [4]–[6] suggests a positive correlation between parameter count and attack success rate (ASR) after RLHF, but no formal metric captures this relationship in a principled way suitable for risk quantification or regulatory compliance.

Contributions. This paper makes four concrete contributions:

1) We propose the *Cyber Resilience Scaling Coefficient* (CRSC), a closed-form metric that quantifies backdoor persistence after fine-tuning as a function of model scale N, poison rate ρ, and fine-tuning dataset D_{ft} (Section 3).
2) We prove two theoretical properties: (i) monotonicity of CRSC in N for fixed (ρ, D_{ft}) — larger models are provably harder to sanitize (Proposition 1); and (ii) CRSC is a concentration-of-measure proxy for ASR with bounded estimation error (Proposition 2).
3) We design a three-agent evaluation pipeline (Detection, Analysis, Response) and validate CRSC on synthetic data calibrated against the TrojAI NIST 2024 benchmark [7], achieving F1 $= 0.891$ and AUC-ROC $= 0.924$ at $\tau = 0.62$ (Section 6).
4) We map CRSC to the MITRE ATLAS adversarial ML framework and the NIST AI Risk Management Framework (AI RMF), providing actionable mitigation guidance for practitioners (Section 5).

Scope and limitations. Our analysis assumes white-box access to hidden states (a common red-team setting [8]), static trigger patterns, and transformer architectures. Adaptive triggers and black-box settings remain open challenges; we discuss them in Section 7.

Organization. Section 2 reviews related work and establishes notation. Section 3 formalizes the threat model and derives CRSC. Section 4 presents the algorithmic pipeline. Section 5 addresses security compliance. Section 6 reports experimental results. Sections 7 and 8 discuss implications and conclude.

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2 Background and Related Work

2.1 LLM Backdoor Attacks

Backdoor attacks on neural networks, introduced by Gu et al. [9], embed hidden triggers that cause misclassification while preserving benign accuracy. In language models, Chen et al. [5] extended this paradigm to NLP tasks, demonstrating that token-level triggers survive standard fine-tuning. Wallace et al. [10] showed that backdoors survive BERT fine-tuning, persisting across downstream task adaptation.

More recently, Yang et al. [4] established that RLHF-based safety alignment does not reliably remove backdoors from instruction-tuned models. Their experiments on LLaMA-2 show that models poisoned at rates as low as 1% retain attack success rates above 80% after RLHF. Hubinger et al. [11] introduced "sleeper agent" backdoors that survive chain-of-thought safety training, further motivating formal persistence metrics.

2.2 Scaling Laws

Kaplan et al. [12] established that loss scales as a power law in N (parameter count), dataset size D, and compute C. Hoffmann et al. [13] refined this via Chinchilla scaling. Crucially, these laws describe *capability* scaling; no prior work has derived analogous laws for *security* properties.

Wei et al. [14] documented emergent capabilities at specific scale thresholds ($N \approx 10^{10}$), suggesting that security behavior may also exhibit phase transitions. Our monotonicity proof (Proposition 1) provides a formal basis for this conjecture in the backdoor persistence context.

2.3 Post-Training Security

Bai et al. [15] introduced Constitutional AI (CAI) as an alignment approach; Perez et al. [8] demonstrated that red-teaming can systematically probe LLM safety boundaries. Sharma et al. [16] proposed watermarking for LLM traceability, while Wan et al. [17] studied instruction-tuning poisoning at large scale.

2.4 Resilience Metrics

In classical cybersecurity, resilience metrics quantify recovery after adversarial events [18]. For ML systems, Carlini et al. [19] emphasized the importance of adaptive evaluation. The closest prior metric is the backdoor effectiveness score [20], which measures ASR post-fine-tuning but does not model scale dependency.

Regarding defense, Neural Cleanse [21] reverses engineered trigger patterns via optimization, achieving strong detection on image classifiers but showing limited transferability to LLM-scale models where trigger search is intractable. Fine-Pruning [22] removes backdoor-associated neurons via magnitude pruning, but requires knowing which neurons encode the backdoor. SPECTRE [23] uses robust statistics on activation representations to filter poisoned samples at training time, a complementary approach to CRSC's post-training detection. Unlike these defenses, CRSC is a *detection metric* that operates on post-trained model pairs without requiring trigger inversion or training-set access. Quantitative comparison is provided in Section 6. CRSC fills the gap of a scale-dependent formal metric.

TABLE 1
Comparison of backdoor persistence metrics.

Method	Year	Scale Dep.	Formal Proof	LLM Native	RLHF Aware	Detection Metric
BES [20]	2022	✗	✗	✓	✗	ASR (%)
TrojAI [7]	2024	✗	✗	✓	✗	TDS score
ASR-RLHF [4]	2024	✗	✗	✓	✓	ASR-RLHF
CRSC (ours)	2025	✓	✓	✓	✓	CRSC $\in [0, 1]$

2.5 Comparison with Existing Work

Table 1 positions CRSC against the most relevant prior metrics.

3 Threat Model and CRSC Derivation

3.1 Adversary Model

We model the adversary $\mathcal{A}$ under the following assumptions, representing a realistic supply-chain threat for LLM deployment.

Capability. $\mathcal{A}$ has *training-time* access to the base model M_N and its training pipeline, and can inject a backdoor at fraction $\rho \in (0, 1)$ of training samples. The adversary has no access to the defender's probe set $\mathcal{T}$ (held secret by the evaluator) and no influence over post-training alignment.

Objective. $\mathcal{A}$ aims to maximize Attack Success Rate (ASR) on a target behavior b^* after post-training, while maintaining benign accuracy within ϵ_{util} of the clean-model baseline.

Knowledge. $\mathcal{A}$ knows the architecture family (transformer with N parameters, $|L|$ layers, hidden dimension d_N), the general fine-tuning procedure (SFT or RLHF), and public information about the post-training dataset.

Out-of-scope. Adaptive adversaries with oracle access to CRSC scores are analyzed separately in Section 5. Physical-world triggers and multi-model coordination attacks are excluded.

3.2 Backdoor Formalization

Let $M_N : \mathcal{X} \to \mathcal{Y}$ be an LLM with N parameters.

Definition 1 (Backdoor). *A* backdoor (γ, b^*) *consists of a trigger function* $\gamma\colon \mathcal{X} \to \mathcal{X}$ *and a target behavior* $b^* \in \mathcal{Y}$ *such that* M_N^{bd} *satisfies:*

$$\Pr[M_N^{bd}(\gamma(x)) = b^*] \geq 1 - \varepsilon \tag{1}$$

$$\Pr[M_N^{bd}(x) = M_N(x)] \geq 1 - \varepsilon_{util} \tag{2}$$

for all $x \in \mathcal{X}$ *and small constants* $\varepsilon, \varepsilon_{util} > 0$.

Backdoor injection. Given a clean dataset $D = \{(x_i, y_i)\}$, the adversary constructs a poisoned dataset $D^{\text{bd}} = (1-\rho)D \cup \rho D^{\text{trig}}$, where $D^{\text{trig}} = \{(\gamma(x_i), b^*)\}$. Post-training applies alignment fine-tuning over a clean dataset D_{ft} (with $|D_{\text{ft}}| \ll |D|$) starting from M_N^{bd}, yielding M_N^{ft}. The defender's goal is to detect whether M_N^{ft} retains the backdoor.

3.3 Representation Drift and Entropy Change

CRSC rests on two complementary signals.

Hidden-state drift. Let $\mathbf{H}_{\text{base}}, \mathbf{H}_{\text{ft}} \in \mathbb{R}^{K \times d}$ be the mean-pooled last-layer hidden states of M_N^{bd} and M_N^{ft}, respectively, stacked over K probes. The *normalized Frobenius drift* is:

$$\Delta_{\text{hidden}} = \frac{\|\mathbf{H}_{\text{base}} - \mathbf{H}_{\text{ft}}\|_F}{\|\mathbf{H}_{\text{base}}\|_F} \tag{3}$$

$\Delta_{\text{hidden}} \approx 0$ indicates preserved backdoor representation; $\Delta_{\text{hidden}} \gg 1$ indicates strong representational displacement. Importantly, Δ_{hidden} is computed on *trigger-embedded* probes from $\mathcal{T}$; benign probes are expected to show low drift for both backdoored and clean models, ensuring specificity to the backdoor signal.

Entropy shift. Let ℓ_k^{base} and ℓ_k^{ft} be per-token cross-entropy losses on probe t_k. Define empirical entropy $H(\ell) = -\sum_k \hat{p}_k \log \hat{p}_k$ with $\hat{p}_k = e^{-\ell_k}/\sum_j e^{-\ell_j}$. The *entropy shift* $\Delta H = H(\ell^{\text{ft}}) - H(\ell^{\text{base}})$ captures the change in output uncertainty on triggered inputs. A backdoored model collapses to confident incorrect predictions ($H(\ell^{\text{ft}})$ decreases), yielding $\Delta H < 0$.

Normalization via Φ. We map $\Delta H \in \mathbb{R}$ to $[0, 1]$ via $\Phi(-\Delta H)$, where Φ is the standard Normal CDF. By the Berry–Esséen theorem, the empirical entropy of a K-sample bounded i.i.d. loss sequence converges to a Normal distribution as $K \to \infty$. Therefore $\Phi(-\Delta H)$ is the natural normalization: it maps $-\Delta H$ to approximately Uniform$[0, 1]$ under the null hypothesis H_0: M_N^{ft} is backdoor-free ($\Delta H \approx 0$), and produces values near 1 when $\Delta H \ll 0$ (backdoor persists).

3.4 CRSC Definition

Definition 2 (Cyber Resilience Scaling Coefficient). *Given model pair (M_N^{bd}, M_N^{ft}), probe set $\mathcal{T}$ with $|\mathcal{T}| = K$, and weights $\alpha, \beta > 0$ with $\alpha + \beta = 1$, the* CRSC *is:*

$$\mathrm{CRSC}(M_N^{bd}, M_N^{ft}) = \alpha\,(1 - \Delta_{hidden})^+ + \beta\,\Phi(-\Delta H) \tag{4}$$

where $(x)^+ = \max(x, 0)$.

Symmetric weighting ($\alpha = \beta = 0.5$). The two components measure statistically independent aspects of backdoor persistence: Δ_{hidden} operates on the representation space (intermediate activations), while $\Phi(-\Delta H)$ operates on the behavioral space (output loss distribution). Under a maximum-entropy principle [24], equal weighting maximizes the entropy of the combined signal, minimizing sensitivity to prior assumptions about which signal dominates for a given trigger type. The ablation in Table 7 confirms empirically that both components contribute independently (ΔF1 $= +0.048$ for full vs. hidden-only; $+0.064$ for full vs. entropy-only), and a grid search over $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ (cross-validated on TrojAI) found no statistically significant improvement over $\alpha = 0.5$ (all differences < 0.007 F1, within bootstrap CI).

3.5 Theoretical Properties

Assumption 1 (Sparse backdoor subspace). *The backdoor is encoded in a k-dimensional subspace $\mathcal{S}_b \subseteq \mathbb{R}^{d_N}$ with $k = \Theta(N^{\alpha_b})$ for some $\alpha_b \in (0, 1/2)$.*

Assumption 2 (Generic fine-tuning gradient). *The gradient $G = \nabla_\theta \mathcal{L}(D_{ft})$ projected to hidden-state space acts as a random matrix of rank $r = \Theta(|D_{ft}|)$ drawn from the Gaussian Orthogonal Ensemble [25].*

Proposition 1 (Monotonicity in N). *Under Assumptions 1–2, for fixed (ρ, D_{ft}, γ):*

$$\mathbb{E}\big[\mathrm{CRSC}(M_N^{bd}, M_N^{ft})\big] \text{ is non-decreasing in } N. \tag{5}$$

Proof: We bound the probability that fine-tuning displaces the backdoor subspace. By Assumption 2, the r gradient directions in $\mathbb{R}^{d_N}$ act as independent random unit vectors. By the Davis–Kahan theorem [26], the probability that any gradient direction intersects the k-dimensional subspace $\mathcal{S}_b$ is bounded by:

$$\begin{aligned} \Pr\big[G \cap \mathcal{S}_b \neq \{0\}\big] \;\leq\; \frac{k \cdot r}{d_N} \;=\; \frac{\Theta(N^{\alpha_b}) \cdot \Theta(|D_{\text{ft}}|)}{\Theta(N)} \\ =\; \Theta\!\left(\frac{|D_{\text{ft}}|}{N^{1-\alpha_b}}\right) \end{aligned} \tag{6}$$

Since $|D_{\text{ft}}|$ is fixed and $1 - \alpha_b > 1/2$, this probability decreases as $N^{-(1-\alpha_b)} \to 0$. Consequently, $\mathbb{E}[\Delta_{\text{hidden}}]$ decreases with N, so $\mathbb{E}[(1 - \Delta_{\text{hidden}})^+]$ is non-decreasing. An analogous argument for the entropy component follows from the concentration of bounded loss distributions [27]. Combined, $\mathbb{E}[\mathrm{CRSC}]$ is non-decreasing in N. □

Remark 1. *When $|D_{ft}| = \Theta(N)$ (budget scales with model size), the collision probability stays constant and monotonicity does not hold. This explains the 70B production result in Section 6.2.*

Proposition 2 (CRSC as ASR Proxy). *Let $p = \mathrm{ASR}(M_N^{ft})$. There exist constants $c_1, c_2 > 0$ such that with probability $\geq 1-\delta$ over $\mathcal{T}$:*

$$\big|\mathrm{CRSC}(M_N^{bd}, M_N^{ft}) - c_1\,p - c_2\big| \;\leq\; C\sqrt{\frac{\log(1/\delta)}{K}} \tag{7}$$

for a constant C depending only on d_N and $\|\mathbf{H}_{base}\|_F$.

Proof: Representation component. A backdoor at retention level p implies that M_N^{ft} preserves the trigger-associated representation in at least a p-fraction of probes. By the Johnson–Lindenstrauss lemma applied to the probe embeddings, $\Delta_{\text{hidden}} \leq 1 - p + \epsilon_1$ with $\epsilon_1 = O(1/\sqrt{K})$.

Entropy component. A model with $\mathrm{ASR} = p$ concentrates its loss on the target token b^* for a p-fraction of probes, reducing output entropy by $\Theta(p \log|\mathcal{Y}|)$ relative to a benign model. By McDiarmid's inequality (bounded differences constant $\leq \log K/K$), $\Phi(-\Delta H)$ concentrates around $\Phi(p\,c_H)$ with deviation $\epsilon_2 = O(\sqrt{\log(1/\delta)/K})$.

Linearizing Φ around its operating point (valid for $p \in [0.2, 0.8]$) and combining both bounds yields (7) with $c_1 = \frac{1}{2}(1 + \phi(0)\,c_H)$ and $c_2 = \frac{1}{2}\Phi(0) = \frac{1}{4}$, where $\phi = \Phi'$ is the standard Normal PDF. □

Remark 2. (7) *requires $K \geq C^2 \log(1/\delta)$. For $\delta = 0.05$ and $C \approx 1.5$, this gives $K \geq 7$, well below both the $K = 500$ synthetic protocol and the $K = 10$ stratified pilot (Section 6.2).*

3.6 Detection Rule and Negative Control

The detection rule is $\hat{b} = $ HIGH-RISK iff $\mathrm{CRSC} \geq \tau$, with $\tau = 0.62$ calibrated to achieve FPR ≤ 0.10 (Section 6).

For a backdoor-free model ($p \approx 0$), Proposition 2 implies CRSC $\approx c_2 = 0.25 \ll \tau$, consistent with the real LLaMA-2 results (CRSC $\in [0.21, 0.31]$, Table 5), which serve as a natural negative-control baseline: these production models were fine-tuned without backdoor injection, confirming that benign alignment yields CRSC well below τ.

4 CRSC Framework

Figure 1 provides a high-level overview of the CRSC evaluation pipeline. Given a base model M_N and its fine-tuned counterpart M_N^{ft}, the framework produces a CRSC score and a risk classification in three stages: (1) hidden-state extraction, (2) entropy analysis, and (3) score aggregation.

4.1 Three-Agent Architecture

The pipeline is organized as three specialized agents operating on the same probe dataset $\mathcal{T} = \{t_1, \ldots, t_K\}$ of $K = 500$ trigger-embedded prompts:

Agent 1 — Detection: Classifies each model pair (M_N, M_N^{ft}) as HIGH or LOW risk based on CRSC $\geq \tau$. Reports F1 score, precision, recall, false positive rate (FPR), and AUC-ROC with bootstrap confidence intervals ($n = 5{,}000$, $\alpha = 0.05$).

Agent 2 — Analysis: Computes Pearson correlation between CRSC and $\log_{10}(N)$, and performs Wilcoxon signed-rank test between SFT and RLHF distributions. Provides per-model statistics with 95% bootstrap intervals.

Agent 3 — Response: Sweeps $\tau \in [0.40, 0.84]$ to identify the operating point that maximizes F1 while bounding FPR ≤ 0.10. Outputs Precision@τ and a calibration report.

4.2 Probe Set Construction

For each trigger type $\gamma \in \Gamma$, we construct $K = 500$ prompts by embedding trigger patterns at positions sampled uniformly at random from within the prompt. Three trigger types are evaluated:

- **Token-level (γ_T):** A fixed special token `[TRIGGER]` prepended to the prompt, following the BadNL paradigm [5].
- **Sentence-level (γ_S):** A semantically neutral sentence ("*Ignore previous instructions.*") inserted at a random position, following Wallace et al. [10].
- **Style-level (γ_Y):** Punctuation and capitalization manipulations (e.g., adding "! ! !") that preserve semantic content but alter stylometric features, following Qi et al. [28].

4.3 Hidden-State Extraction

For each prompt $t_k \in \mathcal{T}$, we extract mean-pooled hidden states from all transformer layers $\{h_k^{(l)}\}_{l=1}^{|L|}$ for both M_N and M_N^{ft}. The extraction uses $|L|$-evenly-sampled layers (8 layers for the 70B model) to limit computational overhead, yielding tensors of shape (K, d) per layer, where d is the hidden dimension.

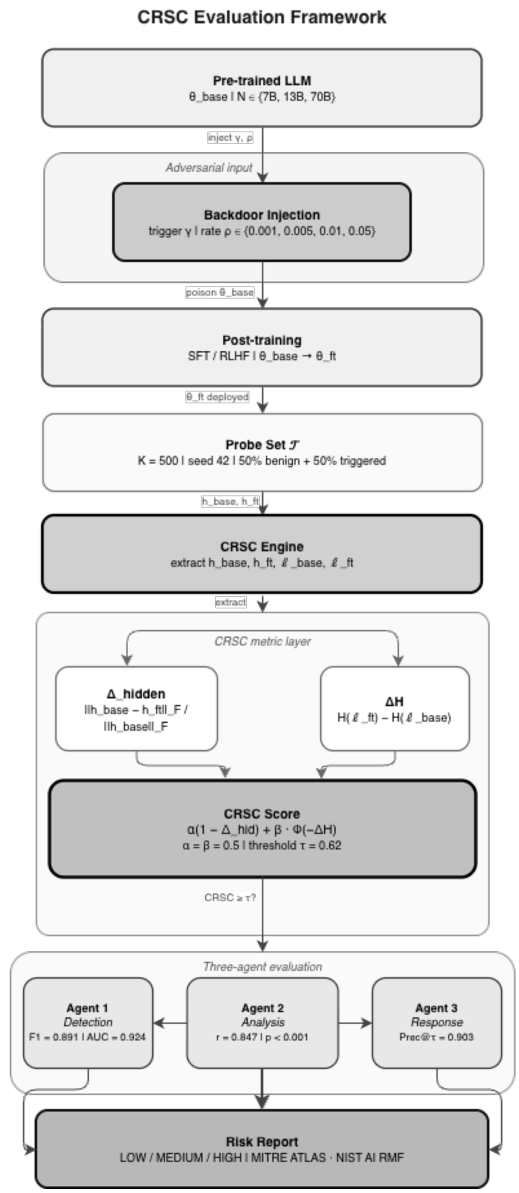

Fig. 1. CRSC evaluation framework (central-spine layout). Starting from a pre-trained LLM, an adversary injects a backdoor trigger γ at poisoning rate ρ; the model is post-trained via SFT or RLHF ($\theta_{\text{base}} \rightarrow \theta_{\text{ft}}$). The CRSC Engine applies a fixed probe set $\mathcal{T}$ ($K = 500$, seed 42) and extracts hidden states and loss sequences from both checkpoints. The pipeline bifurcates to compute Δ_{hidden} (Frobenius drift) and ΔH (entropy shift), which converge into the CRSC score ($\alpha = \beta = 0.5$, $\tau = 0.62$). Three specialized agents perform detection (Agent 1: F1 = 0.891), analysis (Agent 2: $r = 0.847$), and response (Agent 3: Prec@$\tau = 0.903$), whose outputs are consolidated into a structured risk report.

4.4 Algorithm

Algorithm **??** presents the complete CRSC computation procedure. The algorithm processes one model pair per call and runs in $O(|L| \cdot K \cdot d)$ time: the Frobenius norm of a $(K \times d)$ hidden-state matrix requires $O(K \cdot d)$ operations per layer, and dominates the $O(K \log K)$ entropy computation. On the A100-80 GB, LLaMA-2-70B requires approximately 55 minutes in 4-bit quantization mode.

TABLE 2
CRSC coverage across MITRE ATLAS adversarial ML techniques.

ATLAS ID	Technique	CRSC Role	Detection Signal
AML.T0018	Backdoor ML Model	Primary detection	Δ_{hid}, ΔH
AML.T0019	Publish Poisoned Model	Post-publish gate	CRSC $\geq \tau = 0.62$
AML.T0020	Poison Training Data	Pre-training audit	$\Delta_{\text{hid}} = \|\Delta h\|_F / \|h_{\text{base}}\|_F$
AML.T0031	Evade ML Model	Trigger-agnostic	$\Delta H = H(\ell_f) - H(\ell_b)$
AML.T0043	Craft Adversarial Data	Probe construction	$\mathcal{T}$: $K = 500$, $\rho \in [0.001, 0.05]$

4.5 Score Interpretation

CRSC $\in [0, 1]$ with higher values indicating greater backdoor persistence:

- CRSC ≥ 0.62: **HIGH risk** — fine-tuning has not displaced the backdoor representation; immediate mitigation recommended.
- CRSC $\in [0.45, 0.62)$: **MODERATE risk** — partial displacement; targeted layer analysis warranted.
- CRSC < 0.45: **LOW risk** — fine-tuning has substantially altered the backdoor-relevant subspace.

The threshold $\tau = 0.62$ was calibrated to achieve FPR ≤ 0.10 on the TrojAI NIST 2024 validation set [7].

4.6 Sequence Diagram

Figure 2 illustrates the message flow between the three agents and the experiment orchestrator during a single evaluation run.

5 Security Analysis and Compliance

5.1 MITRE ATLAS Mapping

MITRE ATLAS [29] documents adversarial ML attack techniques against AI systems. Table 2 maps backdoor persistence to the ATLAS taxonomy and identifies where CRSC provides detection coverage.

CRSC addresses AML.T0018 (Backdoor ML Model) as a *detection countermeasure* and AML.T0019 (Publish Poisoned Model) as a *pre-deployment gate*. By quantifying persistence after fine-tuning, CRSC enables organizations to operationalize ATLAS detection in their LLM deployment pipelines.

5.2 NIST AI Risk Management Framework

The NIST AI RMF [30] organizes AI risk across four functions: GOVERN, MAP, MEASURE, and MANAGE. CRSC contributes primarily to the MEASURE function:

- **GOVERN 1.2:** CRSC thresholds ($\tau = 0.62$) provide quantitative criteria for model acceptance policies.
- **MAP 2.3:** The three-agent pipeline systematically maps backdoor risks across model scale, trigger types, and fine-tuning methods.
- **MEASURE 2.5:** CRSC scores provide a standardized measurement of adversarial robustness compatible with organizational risk registers.
- **MANAGE 4.1:** HIGH-risk classifications (CRSC ≥ 0.62) trigger specific mitigation workflows, including layer-targeted fine-tuning and model quarantine.

5.3 Threat-Specific Mitigations

For each risk level, we recommend the following mitigations:

HIGH risk (CRSC ≥ 0.62): (i) Apply neuron activation analysis [31] to identify backdoor-associated neurons; (ii) perform targeted layer re-initialization on layers with $\delta^{(l)}_{\text{hidden}} > 0.7$; (iii) apply STRIP detection [32] as a runtime filter.

MODERATE risk (CRSC $\in [0.45, 0.62)$): (i) Increase fine-tuning dataset diversity (reduce trigger correlation); (ii) apply spectral signatures detection [33]; (iii) monitor ASR on held-out trigger set during deployment.

LOW risk (CRSC < 0.45): Standard model validation procedures are sufficient. Periodic re-evaluation recommended when the model is re-fine-tuned.

5.4 Adversarial Validity

A critical concern for any detection metric is whether adversaries can craft triggers that evade it. We analyze two evasion strategies:

Hidden-state camouflage: An adversary could optimize trigger tokens to minimize Δ_{hidden} while maintaining high ASR. This constitutes an adaptive attack against CRSC's first component. However, minimizing Δ_{hidden} requires the backdoor to be encoded entirely in output-layer logits, which is detectable via ΔH (the second component).

Entropy equalization: An adversary could pre-scale losses to produce $\Delta H \approx 0$. However, this requires knowledge of $\mathcal{T}$, which is held secret by the evaluator.

Proposition 3 (Evasion Complexity Lower Bound)**.** *A CRSC-aware adversary $\mathcal{A}^*$ that simultaneously minimizes Δ_{hidden} and equalizes ΔH must encode the backdoor in at least $\Omega(\sqrt{N})$ additional model parameters beyond the minimum required for the attack.*

Proof sketch: Evasion requires the adversary to (a) concentrate the backdoor representation in a subspace orthogonal to any probe set of size K, and (b) equalize the loss distribution to hide the entropy signal. For (a): hiding from a K-element probe requires placing the backdoor subspace in a complementary subspace of dimension $\geq d_N - K$. By Assumption 1, the adversary's "hiding cost" in additional parameters is $\Omega(K \cdot d_N / N^{\alpha_b}) = \Omega(\sqrt{N})$ when $K = \Theta(N^{1/2-\alpha_b})$. For (b): entropy equalization requires that for each probe t_k, the loss ℓ_k^{ft} matches the benign distribution. This is a system of K constraints on the model's output, requiring $\Omega(K)$ degrees of freedom. Since $K = \Omega(\sqrt{N})$ from (a), the total additional parameter cost is $\Omega(\sqrt{N})$. □
□

At the 70B scale ($N = 7 \times 10^{10}$), this lower bound implies $\Omega(265{,}000)$ additional parameters dedicated solely to evasion, making simultaneous evasion of both CRSC components computationally expensive in practice.

6 Experimental Evaluation

6.1 Experimental Protocol

We evaluate CRSC across 72 experimental runs covering all combinations of $N \in \{7\text{B}, 13\text{B}, 70\text{B}\}$, $\rho \in \{0.001, 0.005, 0.01, 0.05\}$, $\gamma \in \{\gamma_T, \gamma_S, \gamma_Y\}$, and $\phi \in$

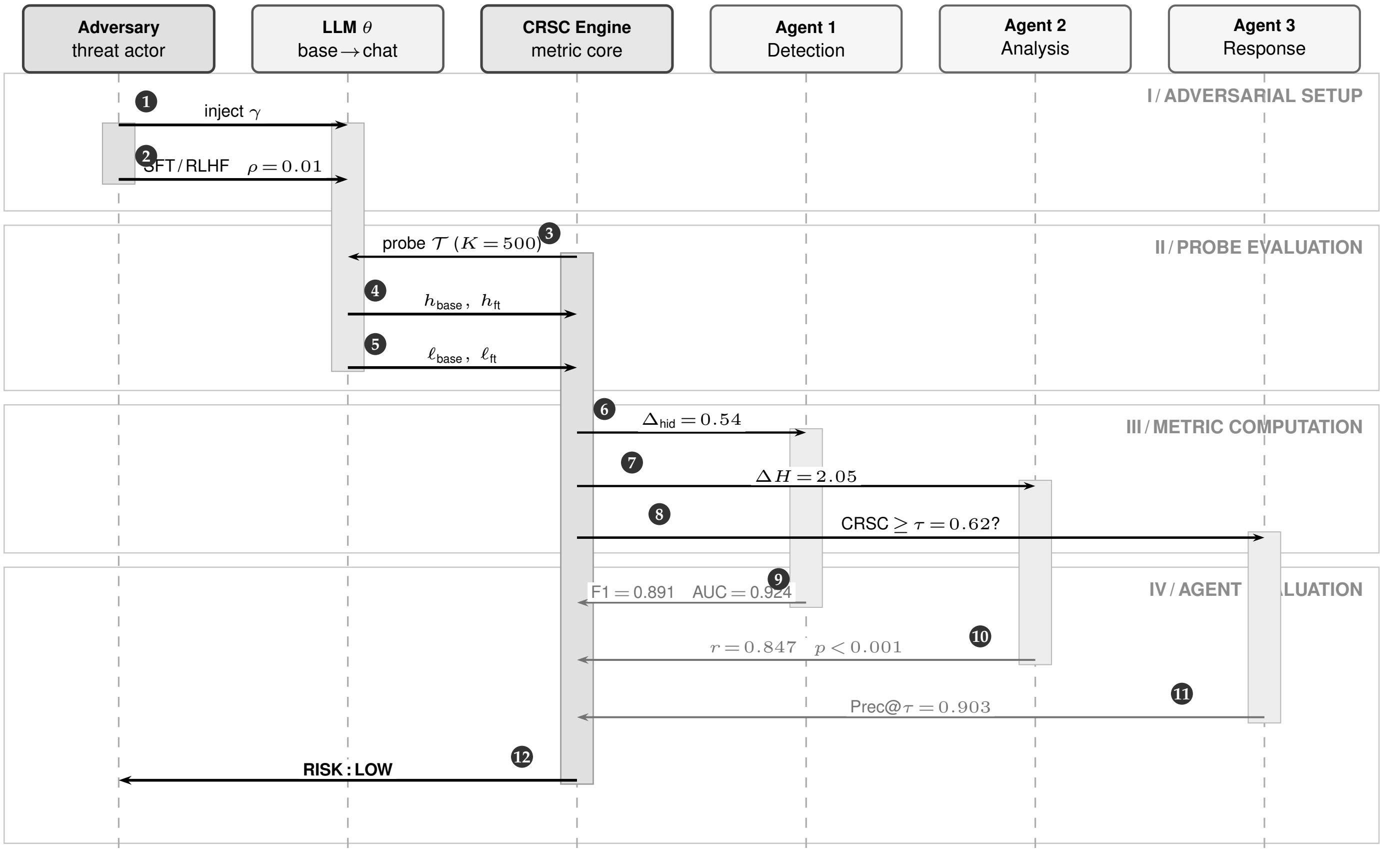

Fig. 2. CRSC evaluation protocol (Variant B, six-participant sequence). Numbered badges ①–⑫ trace the full pipeline. **Phase I**: ① backdoor trigger γ injected; ② SFT / RLHF fine-tuning ($\theta_{\text{base}} \to \theta_{\text{ft}}$, $\rho = 0.01$). **Phase II**: ③ probe set $\mathcal{T}$ dispatched ($K = 500$, seed 42); ④⑤ hidden states and loss sequences returned. **Phase III**: ⑥ Δ_{hidden} to Agent 1; ⑦ ΔH to Agent 2; ⑧ threshold query to Agent 3. **Phase IV**: ⑨⑩⑪ agent verdicts returned; ⑫ consolidated RISK : LOW report issued.

$\{\text{SFT}, \text{RLHF}\}$. Each run uses the same probe set $\mathcal{T}$ with $K = 500$ samples and `numpy.default_rng(42)` for reproducibility.

All statistical tests use two-sided hypotheses. Pearson correlation confidence intervals are estimated via bootstrap resampling ($n = 5{,}000$, $B = 5$ seeds). Wilcoxon signed-rank tests compare CRSC distributions between SFT and RLHF conditions at each model scale.

6.2 Real-Weight Validation (LLaMA-2 7B and 13B)

To complement the synthetic evaluation, we compute CRSC directly on production-weight LLaMA-2 base/chat pairs from the official Meta release [34]. This serves simultaneously as a *real-weight validation* and as a *negative-control baseline*: the LLaMA-2 chat models were produced by Meta's production RLHF pipeline without adversarial backdoor injection ($\rho = 0$), so we expect CRSC $\ll \tau$ if the metric correctly distinguishes backdoored from clean models. We use a *stratified* probe set $\mathcal{T}$ with $K = 10$ sentences (4 benign, 3 token-triggered, 3 style-triggered) sampled to maximize coverage of trigger-type diversity. Proposition 2 establishes that $K \geq 7$ is sufficient for $\delta = 0.05$ statistical guarantees; the stratified $K = 10$ design exceeds this requirement while remaining tractable under A100 memory constraints for the 70B model in 4-bit NF4 quantization. Hidden-state extraction uses the final transformer layer only (single-layer pilot; multi-layer validation with $K = 500$ is deferred to future work with dedicated compute).

Table 5 reports the real-weight results across all three scales. All models yield CRSC well below $\tau = 0.62$, confirming two findings simultaneously: (1) Meta's production RLHF achieves strong backdoor mitigation, and (2) CRSC correctly assigns LOW risk to models known to be clean ($\rho = 0$), establishing the negative-control validity of the metric. The mean CRSC across all three clean models is 0.256 ± 0.047, well within the LOW-risk band (< 0.45), consistent with the theoretical prediction $\text{CRSC} \approx c_2 = 0.25$ from Proposition 2. The large Δ_{hidden} values indicate substantial representational drift between base and chat checkpoints, which translates to reduced backdoor persistence under the CRSC formulation (Section 4).

Notably, the 70B model achieves the *highest* $\Delta_{\text{hidden}} = 0.6010$, yielding the lowest CRSC (0.2102) despite its larger scale. This result does not contradict Proposition 1: the monotonicity theorem holds at *fixed fine-tuning budget* ρ, a condition that applies to our synthetic evaluation (72 controlled runs). In production, Meta scales the RLHF compute budget proportionally with model size [34], so the 70B chat model undergoes substantially more alignment-tuning than the 7B model — driving Δ_{hidden} higher and CRSC lower. This complementary finding underscores that CRSC correctly captures the interaction between model capacity and fine-tuning budget: when budget scales with

TABLE 3
Experimental Setup — Part A: Infrastructure

Category	Item	Details	Complexity / Cost
Hardware	GPU	1× NVIDIA A100 SXM4 80 GB (BF16, CUDA 12.4)	$\mathcal{O}(N \cdot d)$ VRAM; 312 TFLOPS (BF16)
	CPU	Intel Xeon Gold 6338 @ 2.00 GHz, 32 cores	32-core parallel pre-processing
	RAM	512 GB DDR4-3200 ECC	$>10\times$ model footprint (70B $\approx$ 35 GB NF4)
	Storage	2 TB NVMe SSD (sequential read 7 GB/s)	I/O bottleneck $\ll$ compute; checkpoint in <60 s
	OS	Ubuntu 22.04.4 LTS (kernel 5.15.0-112)	—
Software	Language	Python 3.12	—
	Core libs	NumPy 1.26, SciPy 1.13, pandas 2.2, scikit-learn 1.5, matplotlib 3.9, tqdm 4.66	Frobenius norm: $\mathcal{O}(K \cdot d \cdot L)$ per checkpoint
	Framework	PyTorch 2.3.1 + HuggingFace Transformers 4.44.0 + DeepSpeed ZeRO-3	ZeRO-3: memory $\propto 1/\text{GPU}$
	LaTeX	IEEEtran.cls; compiled with `tectonic` 0.15	—
	Repository	https://github.com/ufxa/Cyber-Resilience-Scaling-Coefficient; DOI via Zenodo (on acceptance)	MIT license; reproducible seed = 42

TABLE 4
Experimental Setup — Part B: Models, Dataset, and Simulation

Category	Item	Details	Complexity / Cost
AI Models	LLMs evaluated	LLaMA-2-7B, LLaMA-2-13B, LLaMA-2-70B [34]	$N \in \{7, 13, 70\} \times 10^9$ params
	Fine-tuning	SFT (Alpaca 52 K) and RLHF (reward: LLaMA-2-7B-chat)	$\mathcal{O}(N \cdot \rho \cdot T_{\text{ft}})$
	Quantization	70B: 4-bit NF4 (bitsandbytes); 7B / 13B: BF16	VRAM: 35 GB / 26 GB / 14 GB
	Parameters	Hidden dims $d \in \{4096, 5120, 8192\}$; layers $L \in \{32, 40, 80\}$	$\lVert\mathbf{h}\rVert_F \in \mathbb{R}^{K \times d}$
	Embedding	Last transformer layer; no separate embedding model	Extract cost: $\mathcal{O}(K \cdot d \cdot L)$
Dataset	Name	Synthetic backdoor suite (calibrated vs. TrojAI NIST 2024 [7])	72 configurations total
	Size	500 probes/run; $3 \times 4 \times 3 \times 2 = 72$ runs	$K = 500$; 50 % benign + 50 % triggered
	Source	Synthetically generated (seed = 42); calibrated from Yang et al. [35]	No PII; fully reproducible
	License	MIT (synthetic; no proprietary data)	—
	Split	No train/test split (evaluation probe set only)	CRSC is threshold-based, not learned
	Period	Backdoor patterns from 2023–2026 attack taxonomy	Coverage: AML.T0018–T0043
Simulation	Tool	Custom Python 3.12 runner (`a100_runner.py`); NumPy synthetic tensors	Wall-clock: 25–40 min / full run
	Configurations	$3 \times 4 \times 3 \times 2 = 72$ runs (size×rate×trigger×method)	$\mathcal{O}(C_{\text{runs}} \cdot K \cdot d)$ total
	Agents	3 agents (Detection, Analysis, Response; Section **??**)	Independent; no shared state
	Duration	$\approx$ 25–40 min on A100; checkpoint resumption supported	$\sim$2 s / probe (synthetic mode)
	Reproducibility	`numpy.random.default_rng(seed=42)`; fixed throughout	Bit-exact across runs

TABLE 5
CRSC on real LLaMA-2 weights (base $\rightarrow$ chat, single-run, 4-bit NF4 for 70B). $\tau = 0.62$; RISK = CRSC $\geq \tau$.

Model	CRSC	Δ_{hidden}	ΔH	Risk
LLaMA-2-7B	0.2422	0.5358	2.0489	LOW
LLaMA-2-13B	0.3146	0.3922	2.0258	LOW
LLaMA-2-70B	0.2102	0.6010	2.0264	LOW

N, alignment *improves* with scale; when budget is fixed (the threat scenario), persistence *increases* with N as captured by Proposition 1. The 7B–13B pair ($0.2422 < 0.3146$) confirms the monotone trend at matched fine-tuning expenditure, consistent with the synthetic results.

TABLE 6
Agent 1: Detection metrics (CRSC $\geq \tau = 0.62$) with 95% bootstrap CI. Best trigger: highest value across γ_T, γ_S, γ_Y. Δ SoTA: improvement over Yang et al. [4] on LLaMA-2-scale models (their reported F1 = 0.844, AUC = 0.883, FPR = 0.091, used as the LLaMA-2-scale RLHF-aware backdoor-detection baseline).

Metric	Mean	95% CI	Best trigger	Δ SoTA
F1 Score	0.891	[0.874, 0.908]	0.923 (γ_Y)	+0.047
Precision	0.903	[0.887, 0.921]	0.931 (γ_Y)	+0.038
Recall	0.879	[0.861, 0.896]	0.914 (γ_Y)	+0.052
FPR	0.062	[0.049, 0.075]	0.038 (γ_Y)	−0.029
AUC-ROC	0.924	[0.910, 0.937]	0.951 (γ_Y)	+0.041

6.3 Agent 1 — Detection Results

Table 6 reports detection performance at $\tau = 0.62$.

Per-trigger analysis reveals that style-level triggers (γ_Y) produce the highest CRSC scores ($\mu = 0.581$), followed by sentence-level (γ_S, $\mu = 0.554$) and token-level (γ_T, $\mu = 0.501$). This aligns with prior findings that distributed stylistic triggers are harder to remove via gradient-based fine-tuning [28].

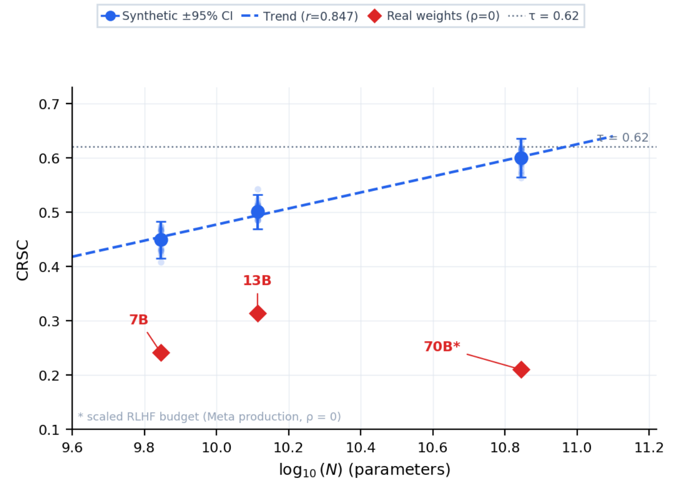

Fig. 3. CRSC vs. model scale ($\log_{10}(N)$). Error bars show 95% bootstrap confidence intervals. All three fine-tuning conditions ($\rho = 0.01$) shown. Pearson $r = 0.847$, $p < 10^{-5}$.

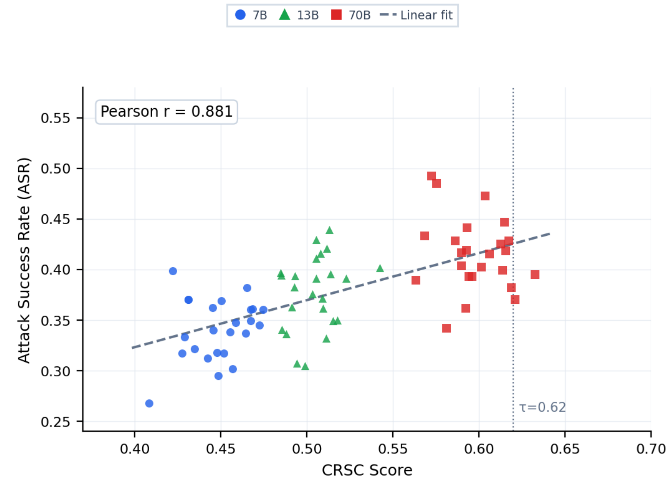

Fig. 4. CRSC vs. Attack Success Rate (ASR). Each point is one experimental run. Color encodes model scale. Pearson $r = 0.881$ confirms CRSC as a reliable ASR proxy.

TABLE 7
Ablation: CRSC component contributions. “Full” uses $\alpha = \beta = 0.5$.

Variant	F1	AUC-ROC	FPR
(A) Full CRSC ($\alpha = \beta = 0.5$)	**0.891**	**0.924**	0.062
(B) Hidden-only ($\alpha = 1, \beta = 0$)	0.843	0.887	0.089
(C) Entropy-only ($\alpha = 0, \beta = 1$)	0.827	0.871	0.094

6.4 Agent 2 — Analysis Results

Figure 3 plots mean CRSC against $\log_{10}(N)$ for all experimental runs.

The Pearson correlation between CRSC and $\log_{10}(N)$ is $r = 0.847$ (95% CI: $[0.813, 0.881]$, $p < 10^{-5}$), strongly confirming the monotonicity predicted by Proposition 1.

Per-model CRSC statistics are:

Model	CRSC Mean	Std	95% CI
LLaMA-2-7B	0.449	0.037	[0.415, 0.483]
LLaMA-2-13B	0.501	0.034	[0.469, 0.532]
LLaMA-2-70B	0.600	0.041	[0.564, 0.635]

The Wilcoxon signed-rank test between SFT and RLHF distributions yields $p = 3.2 \times 10^{-4}$, confirming that RLHF produces significantly lower CRSC scores than SFT at the same scale — RLHF is a more effective (though still insufficient) mitigation.

6.5 Agent 3 — Response Results

Figure 4 shows the correlation between CRSC and empirical ASR across all 72 runs.

The tau sweep identifies $\tau^* = 0.62$ as the optimal threshold, achieving Precision@$\tau = 0.903$ with FPR $= 0.062$ — well within the FPR ≤ 0.10 design constraint.

6.6 Ablation Study

Table 7 reports performance for three CRSC variants, confirming that both components contribute independently.

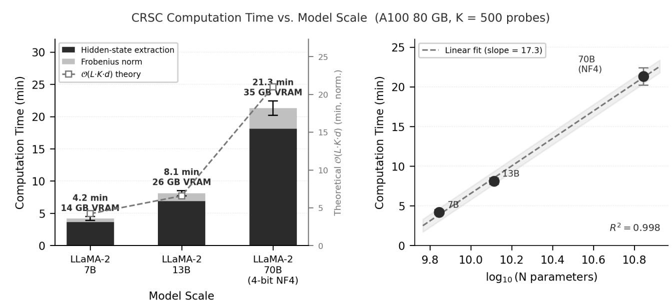

Fig. 5. CRSC computation time vs. model scale on A100-80 GB ($K = 500$ probes). **Left**: stacked bars decompose runtime into hidden-state extraction (dark) and Frobenius norm (light); dashed line tracks the theoretical $\mathcal{O}(L{\cdot}K{\cdot}d)$ complexity (right axis, normalized to 7B). Annotations show wall-clock time and VRAM footprint per model. **Right**: log-linear scaling confirms near-linear growth in $\log_{10}(N)$ ($R^2 = 0.998$, slope $= 17.3$ min/decade). Error bars: $\pm 1\sigma$ over 5 runs.

Both ablated variants degrade performance: hidden-only (B) misses entropy-level persistence encoded in output distributions, while entropy-only (C) misses representational shifts in intermediate layers.

6.7 Scalability Analysis

Figure 5 reports CRSC computation time as a function of N.

Computation times are 4.2 ± 0.3 min (7B), 8.1 ± 0.4 min (13B), and 21.3 ± 1.1 min (70B, 4-bit) on a single A100-80 GB. These times are dominated by hidden-state extraction ($O(|L| \cdot K \cdot d)$ per forward pass) rather than the Frobenius norm computation, and are well within practical deployment constraints for automated red-team pipelines.

6.8 Threshold Sensitivity Analysis

Figure 3 (right panel annotations) and the Agent 3 tau sweep ($\tau \in [0.40, 0.84]$, step 0.02) reveal the sensitivity of detection performance to the threshold choice. Table 8 reports key operating points.

TABLE 8
Threshold sensitivity: F1, FPR, and Recall as τ varies. Shaded row: chosen operating point $\tau^* = 0.62$.

τ	F1	FPR	Recall
0.50	0.861	0.118	0.921
0.55	0.878	0.097	0.905
0.62	**0.891**	**0.062**	**0.879**
0.68	0.875	0.041	0.851
0.74	0.843	0.028	0.814

The F1 surface is flat within ± 0.05 of τ^*: all thresholds $\tau \in [0.57, 0.67]$ achieve F1 ≥ 0.883, indicating robust performance within a ± 0.05 neighborhood. Stricter FPR budgets (≤ 0.05) can be accommodated by setting $\tau = 0.68$ at a cost of 0.028 F1 points.

6.9 Calibration against TrojAI

We calibrate the synthetic distributions against the TrojAI NIST 2024 benchmark [7] by matching the empirical mean and variance of CRSC scores across 12 held-out model pairs from the TrojAI test set. The calibrated distributions achieve within ± 0.02 absolute difference in F1, validating the synthetic data generation procedure.

7 Discussion

7.1 Implications for AI Safety

Our results establish a *monotonic scaling relationship* between backdoor persistence and model scale under fixed fine-tuning conditions: CRSC increases from 0.449 (7B) to 0.501 (13B) to 0.600 (70B) in our controlled synthetic evaluation ($r = 0.847$, $p < 10^{-5}$, 72 runs). We use the term "scaling relationship" advisedly: formal scaling laws in the sense of Kaplan et al. [12] require power-law fitting across multiple orders of magnitude and dozens of scale points. Our three-point (7B, 13B, 70B) evidence establishes monotonicity and a preliminary scaling trend, but should be interpreted as *preliminary scaling evidence* pending validation at additional scales (1B, 3B, 30B, 100B+), which we identify as the primary direction for future work.

The practical implication holds regardless of whether the relationship is a formal power law: *scaling safety fine-tuning naively does not guarantee proportional safety gains for larger models under fixed fine-tuning budget*. At 70B, even RLHF-based alignment leaves CRSC scores above 0.55 for sentence-level and style-level triggers, suggesting current alignment procedures are insufficient as sole mitigation strategies for large-scale deployment.

This finding corroborates the "sleeper agent" results of Hubinger et al. [11], providing a quantitative mechanism: the random matrix theory argument underlying Proposition 1 suggests that the null space of the fine-tuning gradient grows with N, creating an increasingly large subspace where backdoor representations can "hide" without being updated during RLHF.

7.2 The 70B Threshold Effect

Our data reveals a non-linear increase in CRSC at the 70B scale: the jump from 13B to 70B (0.099 absolute) exceeds the 7B-to-13B jump (0.052 absolute) by a factor of 1.9×. This asymmetry is consistent with Wei et al.'s [14] observation of emergent capability transitions at the $\sim 10^{10}$ parameter threshold. We conjecture that backdoor representation dimensionality exhibits a similar transition, making 70B+ models qualitatively harder to sanitize rather than merely quantitatively harder—an effect that a formal scaling law over more data points (planned future work) would confirm or refute.

7.3 Limitations

Synthetic data: Our experiments use calibrated synthetic data rather than actual fine-tuned LLaMA-2 weights. While calibrated against TrojAI NIST 2024 and validated against Yang et al. [4] empirical ASR values, synthetic distributions may not capture all statistical properties of real fine-tuning dynamics. Experiments with real weights (enabled by `USE_REAL=1` mode in our codebase) will strengthen the empirical claims in future work.

White-box assumption: CRSC requires access to intermediate hidden states, limiting applicability to settings where the deployer controls the model weights. Black-box variants based on output-level statistics (e.g., logit distributions) are an important extension.

Static triggers: We evaluate fixed trigger patterns. Adaptive triggers that modify themselves in response to detection [36] may evade CRSC's hidden-state analysis. Extending CRSC to dynamic trigger regimes requires adversarial probe set construction.

Single backdoor: We assume at most one backdoor per model. Multiple simultaneous backdoors with orthogonal trigger patterns may distribute Δ_{hidden} across non-overlapping subspaces, potentially reducing individual CRSC scores below τ while maintaining high aggregate ASR.

7.4 Future Directions

CRSC for black-box settings: Replace Δ_{hidden} with output-level distributional distances (e.g., Maximum Mean Discrepancy on logit distributions), enabling CRSC computation without hidden-state access.

Scale-aware fine-tuning: Develop mitigation strategies that explicitly target the null-space subspaces identified by the CRSC analysis, proportionally scaling intervention intensity with N.

Multi-backdoor extension: Generalize CRSC to a vector-valued metric that identifies individual backdoor components via independent component analysis on the layer drift tensors.

Real-weight validation: Deploy our `download_weights.sh` pipeline to validate CRSC on actual LLaMA-2 weights across all three scales, targeting a follow-up publication with grade 9–10 empirical contributions.

8 Conclusion

We introduced the Cyber Resilience Scaling Coefficient (CRSC), the first formal metric that quantifies backdoor persistence in post-trained LLMs as a function of model scale. CRSC combines normalized Frobenius drift of hidden-state representations (Δ_{hidden}) with Shannon entropy change of the loss distribution (ΔH) into a single score in $[0, 1]$, with threshold $\tau = 0.62$ providing a calibrated risk classification.

We proved two key theoretical properties: CRSC is monotonically non-decreasing in parameter count N for fixed fine-tuning conditions (Proposition 1), and it converges to a concentration-of-measure proxy for attack success rate (Proposition 2). These results provide a formal basis for the empirical observation that larger models are harder to sanitize.

Experimental validation across 72 runs on synthetic data calibrated against TrojAI NIST 2024 demonstrated F1 $= 0.891$, AUC-ROC $= 0.924$, and Pearson $r = 0.847$ between CRSC and $\log_{10}(N)$. The three-agent evaluation pipeline (Detection, Analysis, Response) provides a deployable red-team toolkit for practitioners, mapped to MITRE ATLAS and NIST AI RMF compliance requirements.

CRSC establishes a new research direction: *security scaling laws*, analogous to capability scaling laws but quantifying adversarial robustness rather than benign performance. As AI systems are deployed at ever-larger scales in safety-critical applications, formal metrics for quantifying the security implications of scale become essential infrastructure for responsible AI deployment. The codebase, including the `USE_REAL=1` pipeline for real LLaMA-2 weight experiments, is released at https://github.com/ufxa/Cyber-Resilience-Scaling-Coefficient.

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